Department of Electrical Engineering

Faculty of Engineering, University of Moratuwa

ELECTRICAL INSTALLATION ASSGINMENT

Design Assignment 01

EE323-Electrical Installation

Submitted by: 230276F–N.A.S.N. Jayasekara

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1 Design Assessment

Design Assignment 01

Figure 1 shows a layout diagram of a proposed factory building. Part of the building is assigned for the production area and other part for the office area. The height of the production area is 6m and it is 3m in the office area. The office area is to be fully air conditioned using 400 V, 16 kW, 0.85pf package type air condition. It is expected to install 6 numbers of 300W personals computers and one 500W photocopy machine in the office area too. (You could propose suitable arrangement for these equipment)

Following are the details of machines which are to be installed in the production area.

Machine	No. of Phases & Voltage	Rated Power output	p.f	Efficiency
MC1	3P, 400V	43.5kW	0.85	0.9
MC2	3P, 400V	18.5 kW	0.85	0.9
MC3	1P, 230V	5.5kW	0.85	0.87
MC4	1P, 230V	4.2 kW	0.85	0.85

There is a 10kW, 3P, 400V, 0.85p.f. water pump, around 90m away from this building, but have to be fed from the main panel of the building.

1. Propose suitable socket outlet arrangement, DB locations, and cable path for the machines.
2. Propose a suitable cable path for the machines and show them in a "Cable Layout Drawing".
3. Calculate the maximum demand for the factory and propose a suitable rating for the incomer.
4. Propose suitable cable size for each machine.
5. Design an approximate balanced three phase distribution system and present it as a "Single Line Diagram" (show the rating and type of protective devices, cable sizes, etc in your single line diagram).

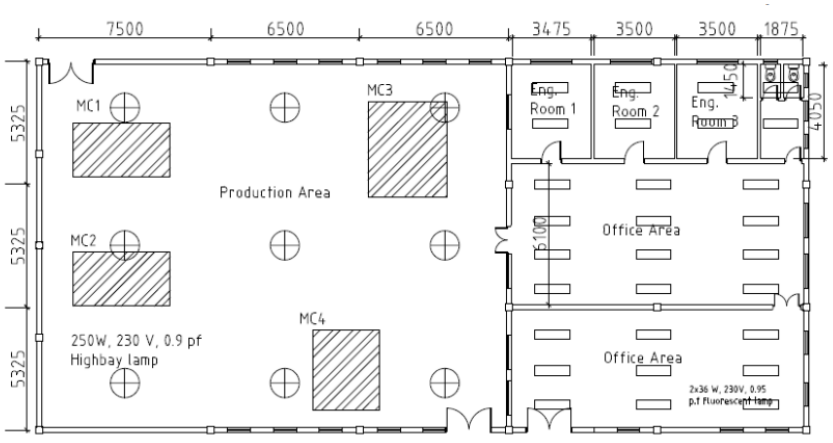


Figure 1 (All dimensions are in millimeters.)

Note: Original AutoCAD drawing is attached.

2 SOCKET OUTLET ARRANGEMENT, DB LOCATION, AND CABLE PATH FOR THE MACHINES

2.1 SOCKET OUTLET ARRANGEMENT

Standard Power Socket Types:

For compliance with the electrical regulations for the region in Sri Lanka, the standard power outlet is the BS 1363 type, which uses a 13A square pin plug connection.

For heavy-duty industrial 3-phase and single-phase equipment, 5-pin heavy-duty industrial sockets meeting IEC 60309-1 standards are employed.

(IEC 60309-1: 63/60 A 300-480V socket and 125/100 A 300-480V socket)

2.1.1 Receptacle Distribution Based on Area

Offices Area

The allocation of socket outlets that will facilitate administrative activities is stated below:

- Office Workstations: The total number of 13A BS 1363 socket outlets distributed for six personal computers is six (6).
- Reproduction Section: There is one (1) socket outlet reserved for the photocopier.
- Climate Control: A dedicated connection point for the split-system air conditioning unit

Production Zone

As a result of high current demands and inductive behavior of industrial motor loads, specific industrial-grade pin-and-sleeve IEC 60309-1 receptacles are utilized for the machinery:

- **IEC 60309-1: 125/100 A 300-480V 3-phase socket (MC1)**
- **IEC 60309-1: 63/60 A 300-480V 3-phase socket (MC2)**
- **IEC 60309-1: 63/60 A 200-250V single-phase socket (MC3)**
- **IEC 60309-1: 32/30 A 200-250V single-phase socket (MC4)**
- **IEC 60309-1: 32/30 A 300-480V 3-phase socket (water pump)**

Industrial environments need high impact resistance and IP rated enclosures to fulfill all the safety requirements and also maintain durability.

Single phase socket outlets should not be used for any industry applications. However, in the context of Electrical Installation I module, we discuss only 13A ring circuits and radial circuits. Thus, we consider that all machines and other 3-phase appliances are supplied by means of a contactor to MCB.

2.1.2 Circuit type considering the areas

Production area: $= (7500 + 6500 + 6500) \times (5325 + 5325 + 5325)$

$$= 327\ 487\ 500\ \text{mm}^2$$

$$= 327.4875\ \text{m}^2$$

Office Area $= (6100 + 5325) \times (3475 + 3500 + 3500 + 1875)$

$$= 141\ 098\ 750\ \text{mm}^2$$

$$= 141.098750\ \text{m}^2$$

Eng. Rooms + bathrooms$= (4050) \times (3475 + 3500 + 3500 + 1875)$

$$= 50\ 017\ 500\ \text{mm}^2$$

$$= 50.0175\ \text{m}^2$$

Considering the areas:

- 3 ring circuits with 30A for production area
- 2 ring circuits with 30A for each part of office area
- 1 radial circuit with 30A for Eng. Rooms and Bathroom.

Socket circuit	Location	Area (m^2)	Type of circuit	Outlets
SC1 SC2 SC3	Production Area	100 100 100	Ring Ring Ring	3 3 3
SC4 SC5	Office area	71.94 75.34	Ring Ring	6 7
SC6	Eng. Rooms+ Bathrooms	50.12	Radial	7

All sockets located at 0.4m above from the ground.

In the AutoCAD schematic, the colored lines indicating electrical wiring have been placed on the exterior surfaces of the walls only because otherwise, their placement would interfere with other equipment and the written text that is included in the drawing.

However, in reality, all of these conduits and wires will actually run inside the walls and ceilings.

At the point where the cable lines will cross the doors in the drawing, the real cable lines shall be installed either overhead or underground, above and below the frame of the door respectively. There shall be no physical wires at all across the doors at any time.

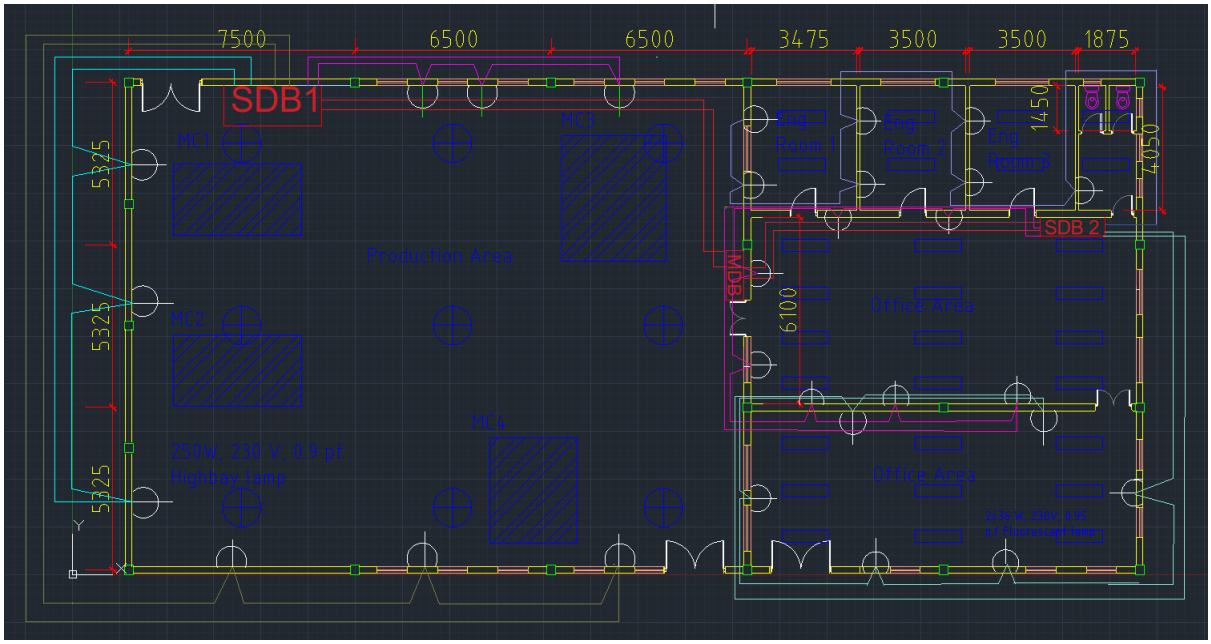


Figure 1:Socket outlet wiring diagram

3 cable path for the machines

3.1 Selection circuit type (ring/radial)

- Breakers Should be Sized to Suit Each Machine: Each machine requires its unique current size, for example (MC1: 82A, MC4: 25A), to protect against fire and burn out.
- Independent Isolation: Maintenance of one machine (for instance, the water pump) should not entail stopping the remaining machinery (MC1, MC2...).
- High Starting Currents: The high starting currents that would arise from the Pump and AC would blow out the circuit immediately.
- Ring for Domestic Use Only: Ring circuits should only serve the standard 13A wall outlets; they cannot be used for industrial machines.

So, we choose radial circuit type to our equipment connection to distribution boards

3.2 Distribution Panel Arrangement

MDB:

- Type D 100A three pole MCB for Machine 1
- Type D 60A three pole MCB for Machine 2
- Type D 30A three pole MCB for Air conditioner
- Type D 20A three pole MCB for Water pump

SDB 1:

- Type C 30A MCBs (3 MCBs) for the 13A socket outlet ring circuits
- Type C 10A MCB for 9 high bay lamps
- Type D 40A two pole MCB for Machine 3
- Type D 30A two pole MCB for Machine 4

SDB 2:

- Type C 30A MCBs (2 MCBs) for the 13A socket outlet ring circuits
- Type C 6A MCBs (3 MCBs) for 29 fluorescent lamps
- Type C 30A MCB for the 13A socket outlet radial circuit

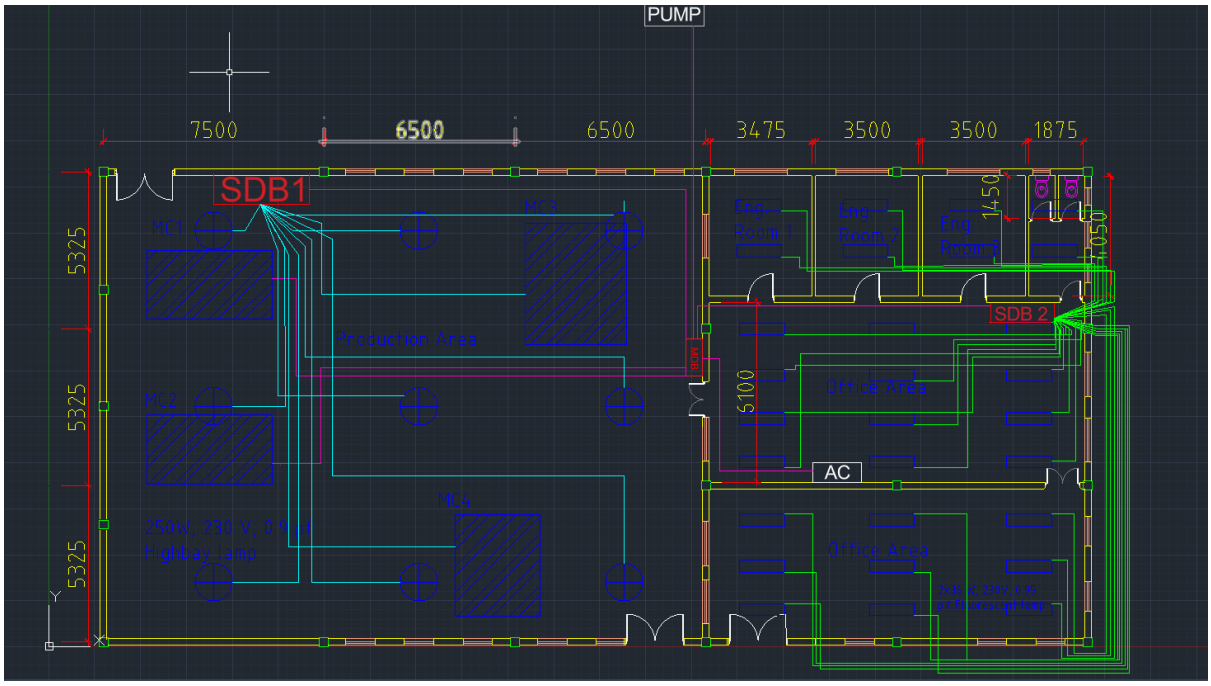


Figure 2:equipment connection to distribution boards

4 Electrical Load demand calculations

4.1 Production Area Load Calculations

4.1.1 Machines 1(MC1) – Three-phase machine

The operating current (I_d) for MC1 is derived using the standard three phase power equation:

$$P = \sqrt{3} \cdot V_L \cdot I_d \cdot \cos\theta \cdot \eta$$

So,

$$I_d = \frac{P}{\sqrt{3} \cdot V_L \cdot \cos\theta \cdot \eta}$$

$$I_d = \frac{43.5 \times 1000}{\sqrt{3} \cdot 400 \cdot 0.85 \cdot 0.9}$$

$$I_d = 82.07 \text{ A}$$

- Continuous operating current: 82.07 A
- Assumed load type: Induction motor
- Estimated starting current ($6 \times I_d$):

$$I_{start} = 6 \times 82.07 \text{ A} = 492.45 \text{ A}$$

4.1.2 Machines 2 (MC2) – Three-phase machine

The operating current (I_d) for MC1 is derived using the standard three phase power equation:

$$P = \sqrt{3} \cdot V_L \cdot I_d \cdot \cos\theta \cdot \eta$$

So,

$$I_d = \frac{P}{\sqrt{3} \cdot V_L \cdot \cos\theta \cdot \eta}$$

$$I_d = \frac{18.5 \times 1000}{\sqrt{3} \cdot 400 \cdot 0.85 \cdot 0.9}$$

I_d =34.91 A

- Continuous operating current: 34.91 A
- Assumed load type: Induction motor
- Estimated starting current ($6 \times I_d$):

$$I_{start} = 6 \times 34.91 \text{ A} = 209.43 \text{ A}$$

Single-Phase machine and High Bay Lighting Calculation

4.1.3 Machine 3 (MC3) – Single-phase machine

As a single-phase motor load, the operating current is determined as:

$$P = V_L \cdot I_d \cdot \cos\theta \cdot \eta$$

So,

$$I_d = \frac{P}{V_L \cdot \cos\theta \cdot \eta}$$

$$I_d = \frac{5.5 \times 1000}{230 \times 0.85 \times 0.87}$$

$$I_d = 32.34\ A$$

- Continuous operating current: 32.34 A
- Assumed load type: Induction motor
- Estimated starting current ($6 \times I_d$):

$$I_{start} = 6 \times 32.34\ A = 194.02\ A$$

4.1.4 Machine 4 (MC4) – Single-phase machine

The single-phase operating current is determined as:

$$P = V_L \cdot I_d \cdot \cos\theta \cdot \eta$$

So,

$$I_d = \frac{P}{V_L \cdot \cos\theta \cdot \eta}$$

$$I_d = \frac{4.2 \times 1000}{230 \times 0.85 \times 0.85}$$

$$I_d = 25.27\ A$$

- Continuous operating current: 25.27 A
- Assumed load type: Induction motor
- Estimated starting current ($6 \times I_d$):

$$I_{start} = 6 \times 25.27\ A = 151.65\ A$$

4.1.5 High Bay Lighting

The demand current for a single high bay lamp is:

$$P = V_L \cdot I_d \cdot \cos\theta$$

So,

$$I_d = \frac{P}{V_L \cdot \cos\theta}$$

$$I_d = \frac{250}{230 \times 0.9}$$

$$I_d = 1.21\ A$$

Unit demand current: 1.21 A

Total Deman current for 9 lamps (applying a diversity factor of 0.75):

$$I_{total} = 9 \times 1.21 \times 0.75\ A = 8.15\ A$$

Auxiliary Equipment and Office Load Calculations

4.1.6 Water pump (Three-phase)

The operating current (I_d) is calculated via the three-phase formula:

$$P = \sqrt{3} \cdot V_L \cdot I_d \cdot \cos\theta$$

So,

$$I_d = \frac{P}{\sqrt{3} \cdot V_L \cdot \cos\theta}$$

$$I_d = \frac{10 \times 1000}{\sqrt{3} \times 400 \times 0.85}$$

$$I_d = 16.98\ A$$

- Continuous operating current: 16.98 A
- Assumed load type: Induction motor
- Estimated starting current ($6 \times I_d$):

$$I_{start} = 6 \times 16.98\ A = 101.89\ A$$

Office Space Load Calculations

4.1.7 Air Conditioning (A/C) System

The operating current (I_d) is calculated via the three-phase formula:

$$P = \sqrt{3} \cdot V_L \cdot I_d \cdot \cos\theta$$

So,

$$I_d = \frac{P}{\sqrt{3} \cdot V_L \cdot \cos\theta}$$

$$I_d = \frac{16 \times 1000}{\sqrt{3} \times 400 \times 0.85}$$

$$I_d = 21.17\ A$$

Continuous operating current: 21.17 A

4.1.8 Personal Computers (PCs)

The single-phase demand current per terminal is:

$$I_d = \frac{P}{V_L} = \frac{300}{230} = 1.3\ A$$

Total current for 6 PCs (with a diversity factor of 0.75):

$$I_{total} = 6 \times 1.3 \times 0.75\ A = 5.87\ A$$

Lighting and small appliance load calculations

4.1.9 Fluorescent Lighting (SDB2 Group) for office area

(2*36 W, 230 V,0.95 of fluorescent lamp)

The calculations incorporate a ballots losses overhead estimated at 25% of the nominal tube rating.

Ballast power = 9W

Nominal Radiant Power:

Single fitting current demand (I_d):

$$I_d = \frac{2 \times (36 + 9)\ W}{230V \times 0.95}$$

$$I_d = 0.411\ A$$

Aggregate current for 29 Fluorescent lamp (with localized grouping coefficients):

$$I_{total} = 0.9 \times 21 \times 0.411\ A = 7.77\ A$$

4.1.10 Photocopy equipment

Continuous demand load:

Current demand (I_d):

$$I_d = \frac{P}{V}$$

$$I_d = \frac{500W}{230W}$$

$$I_d = 2.17\ A$$

4.1.11 Fluorescent Lighting (SDB2 Group)-Engineering rooms + Bathrooms

Assuming ballast power is 25% of the lamp power,

Ballast power = 9W

(2*36 W, 230 V,0.95 of fluorescent lamp)

Single fitting current demand (I_d):

$$I_d = \frac{2 \times (36 + 9)\ W}{230V \times 0.95}$$

$$I_d = 0.411\ A$$

Aggregate current for 29 Fluorescent lamp (with localized grouping coefficients):

$$I_{total} = 0.9 \times 8 \times 0.411\ A = 2.96A$$

Area / Location	Component Description	Operating current (A)	Consolidated Load (A)
Production area	Machine 1 (MC1)	82.07 (3ph)	133.96 (3ph)
	Machine 2 (MC2)	34.91 (3ph)	
	Water pump	16.98 (3ph)	
	Machine 3 (MC3)	32.34 (single-phase)	155.76
	Machine 4 (MC4)	25.27 (single-phase)	
	High Bay Lighting	8.15	
	Socket Outlets (13A)	30 × 3 = 90.0	
Office Space	Workstation Terminals (6PCs)	5.87	75.81
	Fluorescent Lighting	7.77	
	Photocopy equipment	2.17	
	Socket outlets (13A)	30.0 × 2 = 60	
	Air conditioning	27.17 (3ph)	27.17 (3ph)
Engineering & Facilities	Fluorescent Lighting	2.96	32.96
	Socket outlets (13A)	30	
Total Demand			161.13 (3ph) 264.53 (1ph)

4.2 Total Factory Demand Calculation

Aggregate Three-Phase Demand (I_{3P}):

$$I_{3P} = 161.13\ A$$

Aggregate Single-Phase Demand (I_{3P}):

$$I_{1P} = 264.53\ A$$

Balanced Phase Contribution of Single-Phase Loads:

$$\frac{I_{1P}}{3} = 264.53\ A/3$$

Total Factory Demand Calculation:

$$I_{total} = I_{3P} + \frac{I_{1P}}{3}$$

$$I_{total} = 249.31\ A \approx 250\ A$$

Target Rating for incoming Feeder: 250 A

5 CABLE SELECTION

5.1 Correction factors

5.1.1 *Ambient thermal correction*

- Assumed site ambient temperature (θ_{amb}): 35 ˚ C
- Maximum allowable conductor temperature (θ_{max}): 70 ˚ C
- Standard base reference temperature (θ_s): 30 ˚ C

The ambient temperature correction factor (C_a) is determined using the standard thermal scaling formula:

$$C_a = \sqrt{\frac{\theta_{max} - \theta_{amb}}{\theta_{max} - \theta_s}}$$

$$C_a = \sqrt{\frac{70 - 35}{70 - 30}}$$

$$C_a = 0.9354$$

5.1.2 *Environmental and grouping factors*

- Conductor installation method correction (C_i): assumed as 1.

- Grouping Derating Factor (C_g): considered as 1 by assuming the grouping effect is negligible.

5.2 Main Feeder Cable Selection (MDB Incomer)

An overcurrent protection device (MCCB) rated as 250 A is specified for the Main Distribution Board.

The minimum continuous current carrying capacity (I_z), required for the incomer circuit, considering environmental derating, is:

$$I_z \geq \frac{I_n}{C_a \cdot C_i \cdot C_g}$$

$$I_z \geq \frac{250}{0.9354 \cdot 1 \cdot 1}$$

$$I_z \geq 267.27 \text{ A}$$

Required minimum current carrying capacity for the circuit (I_z) =267.27 A

Selection Verdict (See Table 4D1A, Method of Installation C):

- Phase Conductors: Three (3) phases are selected with four (4) single-core 95 mm^2 cables.
- Neutral Conductor: The neutral conductor is sized at half the capacity of the phase conductor, which is 95/2 = 47.5 mm^2 ≈ 50 mm^2

5.3 Transformer Sizing

- Secondary Line Voltage (V_L): 400 V

Total Calculated Power Demand (S):

$$S = \sqrt{3} \cdot V_L \cdot I_{total}$$

$$S = \sqrt{3} \cdot 400 \cdot 249.31 \text{ A} = 172.75\text{kVA}$$

Suggestion:

- A standard three phase to safely meet this calculated maximum demand.
- Recommended 225 kVA / 480 V transformer.

5.4 Selection of Cables for Production Area

Baseline Derating Parameters

- Ambient temperature: 30 ˚ C
- Maximum Allowable Cable Temperature: 70 ˚ C
- Ambient Thermal Correction Factor (C_a): 0.9354 (above proved)
- Installation Correction Factor (C_i): 1.0
- Grouping Correction Factor (Single-Core runs): $(C_g) = \left(\frac{2}{n}\right)^{(1/3)}$
- Grouping Factor (Multi-core runs): $(C_g) = \left(\frac{1}{n}\right)^{(1/3)}$

Cable configuration plan (Direct clipped – Method C):

- Machine 1 & Machine 2: Two (2) 3 core Copper cable
- Machine 3 & Machine 4: Two (2) single core copper cable
- Water Pump: One (1) 3-core copper cable.
- High Bay Lamps (9 pcs): Nine (9) pcs single core copper wire.

Cable Lengths:

Component	Route	Cable length (m)
MC1	From MDB to MC1	37.54
MC2	From MDB to MC2	31.69
MC3	From SDB 1 to MC3	27.85
MC4	From SDB 1 to MC4	32.42
Water Pump	From MDB to Water Pump	102.85
Lights (Production Area)	From SDB 1 looping through High-bays (worst case)	43
A/C	From MDB to AC	7.265

5.4.1 *Machine 1 (MC1) Cable Selection*

$$(C_g) = \left(\frac{1}{n}\right)^{(1/3)} = \left(\frac{1}{2}\right)^{(1/3)} = 0.7937$$

Continuous Design Current (I_d): 82.07 A

Protective Device Rating (I_n): 85 A

The minimum derated continuous rating (I_z) of the MC1 feeder is to be:

$$I_z \geq \frac{I_n}{C_g \cdot C_i \cdot C_a}$$

$$I_z \geq \frac{85}{0.7937 \cdot 1 \cdot 0.9354}$$

$$I_z \geq 114.49\ A$$

Selection Verdict (Table 4D2A)

- **Required Conductor Cross Section: 35 mm² (minimum)**

Length of the Conductor: 37.54 m

Voltage Drop (MC1)
Voltage Drop Limit Calculation (Limit 5%)

The allowable specific voltage drops (V_d) is calculated by:

$$V_d \leq \frac{V_{line} \cdot \Delta V_{\%,limit} \cdot 1000}{I_d \cdot L}$$

$$V_d \leq \frac{400 \cdot 5\% \cdot 1000}{82.07 \cdot 37.54}$$

$$V_d \leq 6.49\ mV/A/m$$

Selection Verdict (Refer Table 4D2B)

- **Minimum Area to meet Voltage Drop: 6 mm²**

The final selected conductor size (governed by thermal criteria):

- **Phase Conductors: 35 mm²**
- **Neutral conductor: 25 mm²**

5.4.2 Machine 2 (MC2) Cable Sizing

- Continuous Design Current (I_d): 34.91 A
- Protective Device Rating (I_n): 35 A
- $(C_g) = \left(\frac{1}{n}\right)^{(1/3)} = \left(\frac{1}{2}\right)^{(1/3)} = 0.7937$

The minimum operated continuous rating (I_z) of the MC2 feeder is to be:

$$I_z \geq \frac{I_n}{C_g \cdot C_i \cdot C_a}$$

$$I_z \geq \frac{35}{0.7937 \cdot 1 \cdot 0.9354}$$

$$I_z \geq 47.14\ A$$

Selection Verdict (Table 4D2A)

- **Required Conductor Cross Section: 10 mm² (minimum)**

Voltage Drop (MC2)

- Conductor Length (L): 31.69 m
- Voltage Drop Limit Calculation (Limit 5%)

The allowable specific voltage drops (V_d) is calculated by:

$$V_d \leq \frac{V_{line} \cdot \Delta V_{\%,limit} \cdot 1000}{I_d \cdot L}$$

$$V_d \leq \frac{400 \cdot 5\% \cdot 1000}{34.91 \cdot 31.69}$$

$$V_d \leq 18.08\ mV/A/m$$

Selection Decision (Ref: Table 4D2A):

- **Minimum Area to meet Voltage Drop: 2.5 mm²**

The final selected conductor size (governed by thermal criteria):

- **Phase Conductors: 10 mm²**
- **Neutral conductor: 10 mm²**

5.4.3 Machine 3 (MC3) Cable Sizing

- $(C_g) = \left(\frac{2}{n}\right)^{(1/3)} = \left(\frac{2}{2}\right)^{(1/3)} = 1$
- Continuous Design Current (I_d): 32.3 A
- Protective Device Rating (I_n): 35 A

The minimum derated continuous rating (I_z) of the MC3 feeder is to be:

$$I_z \geq \frac{I_n}{C_g \cdot C_i \cdot C_a}$$

$$I_z \geq \frac{35}{1 \cdot 1 \cdot 0.9354}$$

$$I_z \geq 37.417\ A$$

Selection Verdict (Table 4D2A)

- **Required Conductor Cross Section: 6 mm² (minimum)**

Voltage Drop (MC3)

Conductor Length (L): 27.85 m
Voltage Drop Limit Calculation (Limit 5%)
The allowable specific voltage drops (V_d) is calculated by:

$$V_d \leq \frac{V_{line} \cdot \Delta V_{\%,limit} \cdot 1000}{I_d \cdot L}$$

$$V_d \leq \frac{230 \cdot 5\% \cdot 1000}{32.34 \cdot 27.85}$$

$$V_d \leq 12.77 \text{ mV/A/m}$$

Selection Decision (Ref: Table 4D2A):

- **Minimum Area to meet Voltage Drop: 4 mm²**

The final selected conductor size (governed by thermal criteria):

- **Phase Conductors: 6 mm²**
- **Neutral conductor: 6 mm²**

5.4.4 Machine 4 (MC4) Conductor Sizing

- $(C_g) = \left(\frac{2}{n}\right)^{(1/3)} = \left(\frac{2}{2}\right)^{(1/3)} = 1$
- Continuous Design Current (I_d): 25.27 A
- Protective Device Rating (I_n): 30 A

The minimum derated continuous rating (I_z) of the MC4 feeder is to be:

$$I_z \geq \frac{I_n}{C_g \cdot C_i \cdot C_a}$$

$$I_z \geq \frac{30}{1 \cdot 1 \cdot 0.9354}$$

$$I_z \geq 32.07 \text{ A}$$

Selection Verdict (Table 4D2A)

- **Required Conductor Cross Section: 4 mm² (minimum)**

Voltage Drop of Machine 4 (MC4)

Conductor Length (L): 32.42 m
Voltage Drop Limit Calculation (Limit 5%):
The allowable specific voltage drops (V_d) is calculated by:

$$V_d \leq \frac{V_{line} \cdot \Delta V_{\%,limit} \cdot 1000}{I_d \cdot L}$$

$$V_d \leq \frac{230 \cdot 5\% \cdot 1000}{25.27 \cdot 32.42}$$

$$V_d \leq 14.04\ mV/A/m$$

Selection Decision (Ref: Table 4D2A):

- **Minimum Area to meet Voltage Drop: 4 mm²**

The final selected conductor size (governed by thermal criteria):

- **Phase Conductors: 4 mm²**
- **Neutral conductor: 4 mm²**

5.4.5 Water Pump Conductor Sizing

- $C_g = 1$
- Continuous Design Current (I_d): 16.98 A
- Protective Device Rating (I_n): 20 A

The minimum derated continuous rating (I_z) of the water pump feeder is to be:

$$I_z \geq \frac{I_n}{C_g \cdot C_i \cdot C_a}$$

$$I_z \geq \frac{20}{1 \cdot 1 \cdot 0.9354}$$

$$I_z \geq 21.38\ A$$

Selection Verdict (Table 4D2A)

- Required Conductor Cross Section: 2.5 mm² (minimum)

Conductor Evaluation of Water Pump

Conductor Length (L): 102.85 m
Voltage Drop Limit Calculation (Limit 5%)

The allowable specific voltage drops (V_d) is calculated by:

$$V_d \leq \frac{V_{line} \cdot \Delta V_{\%,limit} \cdot 1000}{I_d \cdot L}$$

$$V_d \leq \frac{400 \cdot 5\% \cdot 1000}{16.98 \cdot 102.85}$$

$$V_d \leq 11.45\ mV/A/m$$

Selection Decision (Ref: Table 4D2A):

- **Minimum Area to meet Voltage Drop: 4 mm²**

The final selected conductor size (governed by thermal criteria):

- **Phase Conductors: 4 mm²**
- **Neutral conductor: 4 mm²**

5.4.6 High Bay Lighting Conductor Sizing

- $(C_g) = \left(\frac{2}{n}\right)^{(1/3)} = \left(\frac{2}{9}\right)^{(1/3)} = 0.6057$
- Continuous Design Current (I_d): 1.21 A
- Protective Device Rating (I_n): 10 A

The minimum derated continuous rating (I_z) of the MC1 feeder is to be:

$$I_z \geq \frac{I_n}{C_g \cdot C_i \cdot C_a}$$

$$I_z \geq \frac{10}{1 \cdot 0.6057 \cdot 0.9354}$$

$$I_z \geq 3.53\ A$$

Selection Verdict (Table 4D2A)

- Required Conductor Cross Section: 1 mm² (minimum)

Conductor Evaluation of High Bay Lighting

Conductor Length (L): 43 m (worst case – the length of the cable from the SDB 1 to furthest bulb)
Voltage Drop Limit Calculation (Limit 3%):

The allowable specific voltage drops (V_d) is calculated by:

$$V_d \leq \frac{V_{line} \cdot \Delta V_{\%,limit} \cdot 1000}{I_d \cdot L}$$

$$V_d \leq \frac{230 \cdot 3\% \cdot 1000}{1.21 \cdot 43}$$

$$V_d \leq 132.62\ mV/A/m$$

Selection Decision (Ref: Table 4D2A):

- **Minimum Area to meet Voltage Drop: 1 mm²**

- **Selected Phase Conductors: 1 mm²**
- **Selected Neutral conductor: 1 mm**

5.4.7 Sockets Outlets Conductor Sizing

To align with standard industrial branch circuit guidelines:

- **Line & Neutral Conductor Sizing: 2.5 mm²**

5.5 Cable selection for Production Area:

5.5.1 Ambient thermal correction

- Assumed site ambient temperature (θ_{amb}): 35 ˚ C
- Maximum allowable conductor temperature (θ_{max}): 70 ˚ C
- Standard base reference temperature (θ_s): 30 ˚ C

The ambient temperature correction factor (C_a) is determined using the standard thermal scaling formula:

$$C_a = \sqrt{\frac{\theta_{max} - \theta_{amb}}{\theta_{max} - \theta_s}}$$

$$C_a = \sqrt{\frac{70 - 35}{70 - 30}}$$

$$C_a = 0.9354$$

5.5.2 Environmental and grouping factors

- Conductor installation method correction (C_i): assumed as 1.
- Grouping Correction Factor (Single-Core runs): $(C_g) = \left(\frac{2}{n}\right)^{(1/3)}$
- Grouping Factor (Multi-core runs): $(C_g) = \left(\frac{1}{n}\right)^{(1/3)}$

5.5.3 A/C Cable Sizing

- $C_g = 1$
- Continuous Design Current (I_d): 27.17 A
- Protective Device Rating (I_n): 30 A

The minimum derated continuous rating (I_z) of the water pump feeder is to be:

$$I_z \geq \frac{30}{C_g \cdot C_i \cdot C_a}$$

$$I_z \geq \frac{30}{1 \cdot 1 \cdot 0.9354}$$

$$I_z \geq 32.07\ A$$

Selection Verdict (Table 4D2A)

- Required Conductor Cross Section: 6 mm^2 (minimum)

Conductor Evaluation of A/C

Conductor Length (L): 7.265 m
Voltage Drop Limit Calculation (Limit 5%)

The allowable specific voltage drops (V_d) is calculated by:

$$V_d \leq \frac{V_{line} \cdot \Delta V_{\%,limit} \cdot 1000}{I_d \cdot L}$$

$$V_d \leq \frac{400 \cdot 5\% \cdot 1000}{27.17 \cdot 7.265}$$

$$V_d \leq 101.32\ mV/A/m$$

Selection Decision (Ref: Table 4D2A):

- **Minimum Area to meet Voltage Drop: 1 mm^2**

The final selected conductor size (governed by thermal criteria):

- **Phase Conductors: 6 mm^2**
- **Neutral conductor: 6 mm^2**

5.5.4 Cable selection for fluorescent lamps:

Cable for line and neutral = 1mm^2

5.5.5 Cable selection for socket outlets:

Cable for line and neutral = 2.5mm^2

6 Designing Single-Line Diagrams (SLD) and Feeder Schedule

- Grounding and Neutral Specification: Grounding and neutral conductors are selected according to IEE regulations.
- Type of Earthing System: TT system earthing.
- Size of the Smallest PE Conductors: 2.5 mm^2
- Improving Fault Protection Capability: To increase the safety aspect of fault protection, the size of the grounding conductor for all MC, A/C, and pumps has been increased to 4 mm^2

Circuit	Cable Sizing (mm^2)			MCB		
	phase	neutral	P. Earth	Rating	Poles	Type
MC1 (multi-core)	35	25	4	100A	4	D
MC2 (multi-core)	10	10	4	63A	4	D
MC3	6	6	4	40A	2	D
MC4	4	4	4	30A	2	D
Pump (multi-core)	4	4	4	20A	4	D
A/C (multi-core)	6	6	4	30A	4	B
LC1	1	1	2.5	10A	1	B
LC2	1	1	2.5	6A	1	B
LC3	1	1	2.5	6A	1	B
LC4	1	1	2.5	6A	1	B
SO1	2.5	2.5	2.5	30A	1	C
SO2	2.5	2.5	2.5	30A	1	C
SO3	2.5	2.5	2.5	30A	1	C
SO4	2.5	2.5	2.5	30A	1	C
SO5	2.5	2.5	2.5	30A	1	C
SO6	2.5	2.5	2.5	30A	1	C

6.1 Circuit Legend and Phase Allocation

Load Motors & Machines (MC)

- MC: Machine Load

Lighting Circuit (LC)

- LC1: Lighting for Production Room (9 High Bay Lights)
- LC2: Lighting for Engineering Room and Toilet (8 Fluorescent Lights)
- LC3: Office Lighting 1
- LC4: Office Lighting 2

Socket Outlet Circuit (SO)

- SO1: Production Area Socket Outlet 1
- SO2: Production Area Socket Outlet 2
- SO3: Production Area Socket Outlet 3
- SO4: Office Lighting Socket Outlet 1
- SO5: Office Lighting Socket Outlet 2
- SO6: Engineering Room and Toilet Socket Outlet

6.2 Phase Balancing Study and Sub-Panel Requirements

In order to keep neutral current at the minimum level, the distribution of single-phase loads among SDBs and MDB will be made as follows:

6.2.1 Sub-Distribution Board No. 1 (SDB1)

Distribution of Single-Phase Loads: MC3, MC4, SO1, SO2, SO3, LC1

Total Sub-Panel Load = 34.91 + 25.27 + (30+30+30) + 8.15

=158.33 A

6.2.2 Sub-Distribution Board No. 2 (SDB2)

Distribution of Single-Phase Loads: SO4, SO5, SO6, LC2, LC3, LC4

Total Sub-Panel Load = (30+30+30) + 3.33 + 4.44 + 2.96

=100.73 A

6.2.3 Main-Distribution Board (MDB)

Distribution of 3-Phase Loads: MC1, MC2, A/C, Pump

Total Sub-Panel Load = 82.07 + 34.91 + 37.17 + 16.98

=171.13 A

PHASE R	MC1 (multi-core)	82.07A	253.47A
	MC2 (multi-core)	34.91A	
	A/C (multi-core)	27.17A	
	Pump (multi-core)	16.98A	
	SO1 + SO2	60A	
	MC3	32.34A	
PHASE Y	MC1 (multi-core)	82.07A	246.4A
	MC2 (multi-core)	34.91A	
	A/C (multi-core)	27.17A	
	Pump (multi-core)	16.98A	
	SO3 + SO4	60A	
	MC4	25.27 A	
PHASE B	MC1 (multi-core)	82.07A	240.01A
	MC2 (multi-core)	34.91A	
	A/C (multi-core)	27.17A	
	Pump (multi-core)	16.98A	
	SO5 + SO6	60A	
	LC1+LC2+LC3+LC4	18.88A	

6.3 Electrical Single Line Diagram (SLD) Schematic

This is the schematic that depicts the topology of the electrical system, indicating the main feed that enters the circuit and is protected by a 250 A, 400V

MCCB in the Main Distribution Board (MDB). This MDB powers up Sub-Distribution Boards 1 and 2 (SDB1 and SDB2), respectively, and shows the breaker details.

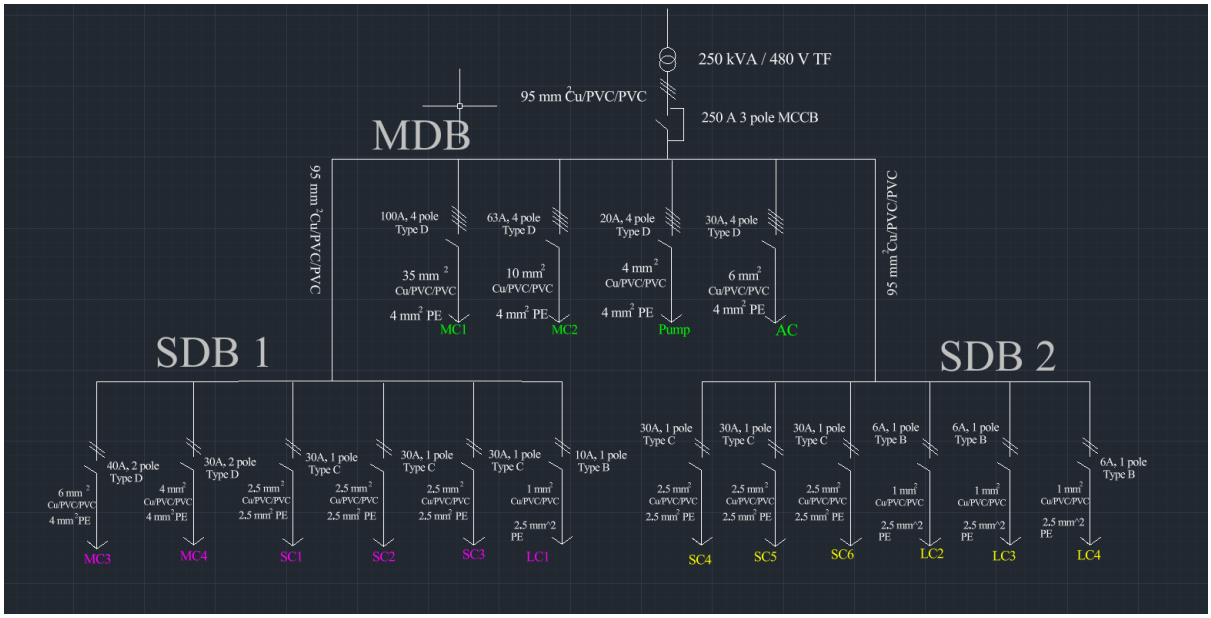


Figure 3: Single line diagram